

BrandPulse Sprint 1

Competitive Vulnerability Intelligence Engine

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Internship Project · August 2026

What BrandPulse Is Now

- Continuously monitors professional news sources and RSS feeds for competitor vulnerabilities
- Detects six vulnerability types: System Outages, Price Increases, PR Crises, Layoffs, Product Defects, Data Breaches
- Calculates an Opportunity Score (0-100) based on severity, coverage volume, and urgency
- Routes intelligence to the correct department: Sales, Marketing, HR, or Product
- Stores AI-generated competitive intelligence in the database and makes it accessible through REST APIs.

Business Value: Turns competitor weaknesses into actionable revenue opportunities — automatically, for \$0 in API costs.

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A competitor's problem is your revenue opportunity.

When Square POS suffers a nationwide outage, BrandPulse tells our client's sales team exactly what to say and who to target — within minutes.

Baseline vs Sprint 2

(Baseline)

Data: Reddit RSS feeds

Content: Short user posts

Validation: Basic deduplication

AI: Sentiment + Intent classifiers (2 models)

Processing: Ingestion → Sentiment + Intent → Database

Output: Charts & tables on a dashboard

Sprint 1 (Now)

Data: TechCrunch, Google News, Reddit

Content: Full professional articles (500-1000 words)

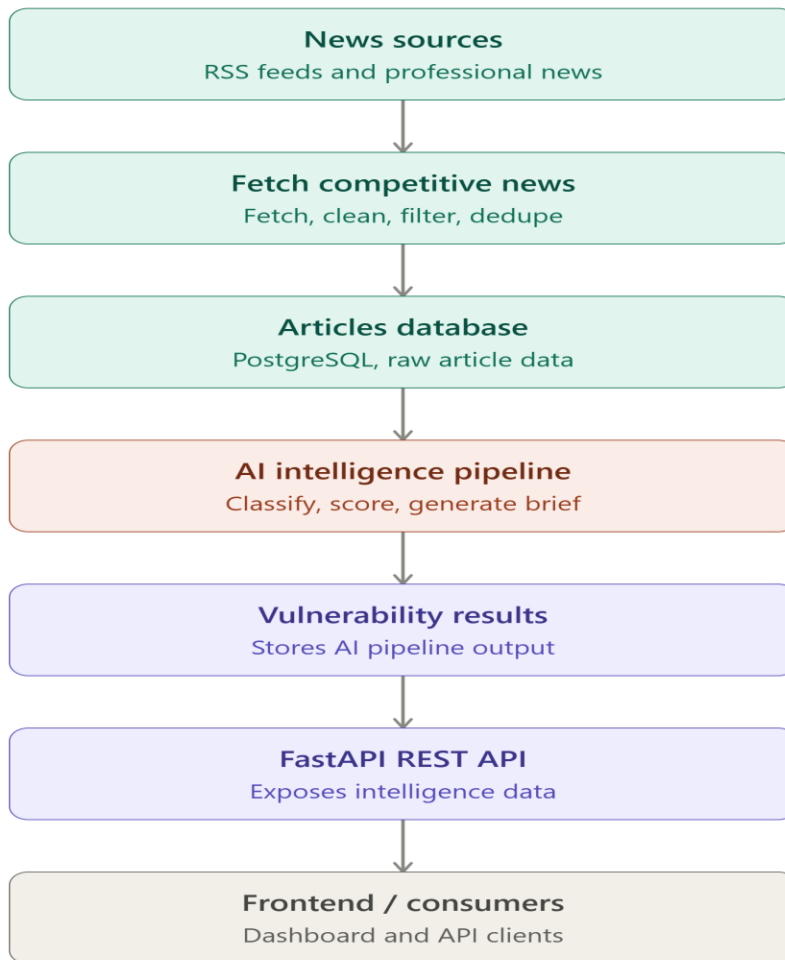
Validation: Competitor + context-based filtering and article persistence

AI: Local 4-stage vulnerability intelligence pipeline

Processing: Automated ingestion → DB → AI analysis → VulnerabilityResult storage

Output: Vulnerability results, Opportunity Scores, Action Briefs & Fact Audit results

Flow Diagram



4-Stage Local AI Pipeline (Basim's Layer)

- ▷ Stage 1 — The Bouncer: cross-encoder/nli-deberta validates relevance, classifies vulnerability type, drops irrelevant articles
- ▷ Stage 2 — The Scorekeeper: pure Python heuristic — Severity (40%), Volume (30%), Urgency (30%) → Opportunity Score 0-100
- ▷ Stage 3 — The Copywriter: Llama-3.2-3B / Qwen-2.5-1.5B generates action brief, routes to department, writes outreach copy
- ▷ Stage 4 — The Fact-Checker: cross-encoder audits generated copy against the source article and flags unsupported claim
- ▷ [AI Pipeline Diagram](#)

Runs entirely on local hardware — no cloud API costs.

Team Contributions

- **Ismail** — Data & Infrastructure Engineer: RSS ingestion from 8+ sources, two-factor validation engine, Docker, frontend routing
- **Basim** — AI/ML Engineer: 4-stage local AI pipeline — Bouncer, Scorekeeper, Copywriter (Llama/Qwen SLMs), Fact-Checker. All models run locally, \$0 cloud cost
- **Momina** — Backend & API Engineer: REST API endpoints, Pydantic schemas, database service layer, APScheduler background jobs, PostgreSQL models (8 tables, 20+ indexes), FastAPI backend

Thanks!

Any questions?

BrandPulse Sprint 1 — Ismail · Basim · Momina